Project Title: Searching for Systematics in the Cosmic Microwave Background with Simulation-Based Inference

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Project summary

When a new paper makes exciting claims about a cosmological or astrophysical process, why should we trust them? A necessary, but not sufficient, piece of evidence is the use of "null and consistency tests"; these are simple tests that are designed to demonstrate that systematic effects are not unduly influencing the results. There is a wealth of Bayesian and Frequentist tools that underpin these tests. Simulation based-inference (SBI) is an exciting new paradigm where the explicit statistical model of classical approaches is replaced by an implicit model comprised of simulations. The goal of this project is to understand how to translate the suite of null and consistency tests to this new framework and to apply these tools to a cosmic microwave background dataset (Fig 1).

Background

Modern cosmological and astrophysical experiments are highly complex and, typically, astrophysicists are looking for subtle signals in these datasets. A key challenge is to understand how to differentiate the signal of interest from an unaccounted for instrumental, observational or computational (i.e. a bug) effect. The current gold standard is to construct an extensive suite of "null and consistency tests"; these tests typically involve dividing the data in parts and asking are these statistically consistent. Each of these tests is designed to isolate a potential source of systematic bias and passing these tests provides confidence that the result is robust to the tested sources of bias. Despite their apparent simplicity, these are proven tools for routing out unexpected biases.

The complexity of modern datasets often means that it is not possible to write down an analytical form for the probability distribution of our observables (consider the distribution of matter in the Universe). The leading approach to address this challenge is simulation-based inference (SBI). In this class of methods, we replace the analytical probability distribution with an implicit distribution formed by a large ensemble of simulations. A further benefit of this approach is that it also that is also directs researcher's time to be focused on understanding the physics of the processes (and implementing that in the simulations) rather than developing complex (and approximate) analytical probability distributions. Machine learning methods are increasingly used to enable fast and precise SBI analyses; however, the black box nature of these methods is a source of concern for many. How can we trust these analyses if we do not "understand" all the intervening steps?

Project details

The goal of this project is to translate the ideas of “null and consistency” tests to the SBI framework. To demonstrate the efficacy of the developed ideas, we will use a set of mock cosmic microwave background (CMB) observations. Null and consistency tests are extensively used in that field — leading analyses include hundreds of them! — and it is simple to generate simulated datasets that contain systematics.

Specifically, the project will have three stages: first we will implement a simple SBI analysis for the cosmic microwave background power spectrum. Second, we will introduce divisions of the data, called splits, with plausible systematics for this analysis and demonstrate how a naïve analysis would miss the systematics and produce a biased inference. Finally, we will implement our SBI-null tests to demonstrate that we can identify the systematic effect. Our starting point for SBI-null tests will be to compute the probability for one data split given a second split, in the SBI framework.

Prerequisite Skills required

The key skills are reasonable programming skills, rudimentary knowledge of cosmology, statistics and machine learning.

Other skills used in the project

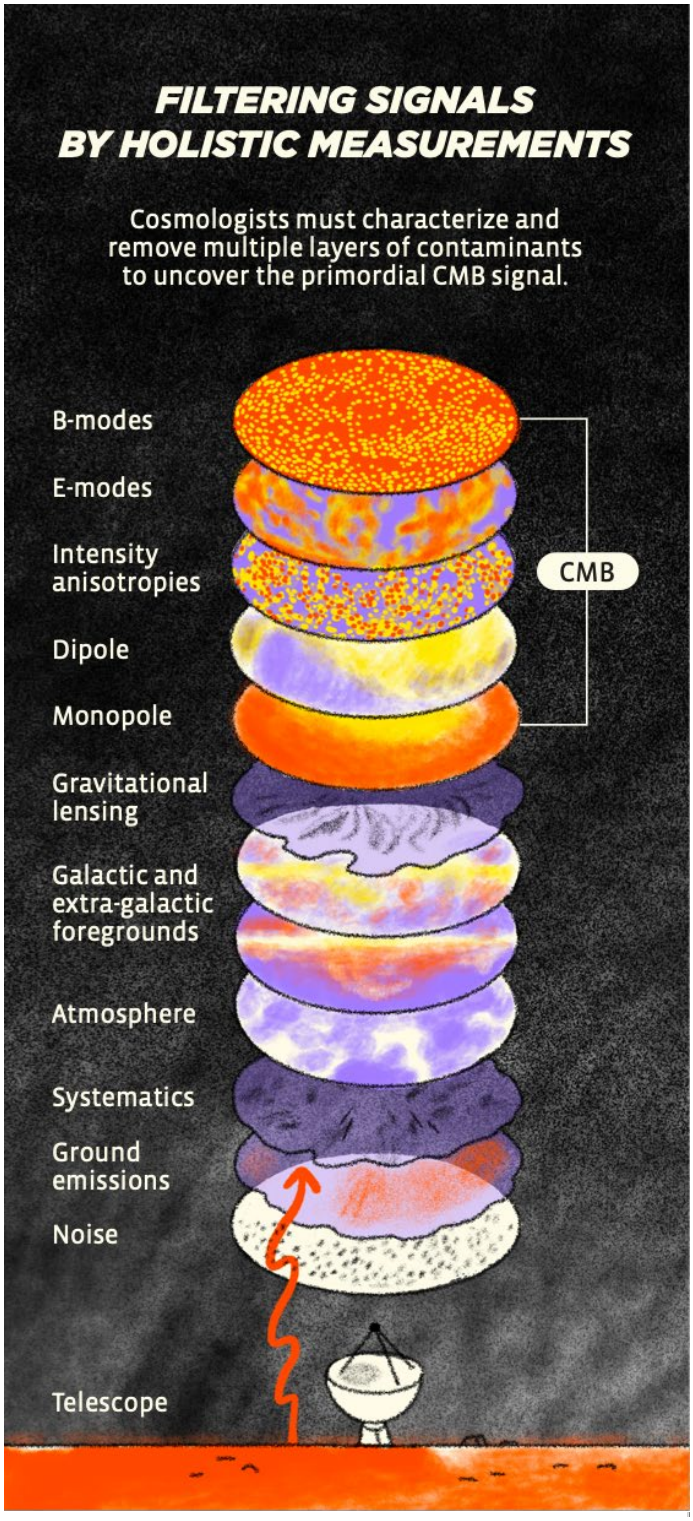
An understanding of Bayesian statistical inference or experience with any ML framework (Tensorflow, pytorch, JAX…) is desirable, but not necessary.

Acceptable Programming Languages

Python

Useful References (List of important papers/review articles relevant to the project)

- The Atacama Cosmology Telescope: DR6 Power Spectra, Likelihoods and ΛCDM Parameters – The ACT collaboration (arXiv: 2503.14452)
- The frontier of simulation-based inference Kyle Cranmer, Johann Brehmer, Giles Louppe (arXiv: 1911.01429)
- Evidence Networks: simple losses for fast, amortized, neural Bayesian model comparison Niall Jeffrey, Ben Wandelt (arXiv: 2305.11241)
- Tests for model misspecification in simulation-based inference: from local distortions to global model checks, Noemi Anau Montel, James Alvey, Christoph Weniger (arXiv: 2412.15100)



Schematic of CMB systematics (credit Josquin Errard)

Figure 1: Schematic of CMB systematics (credit Josquin Errard)